

Douglas Gordon Daly

Principal AI Engineer / Technical Leader

Portland, OR | douglasgdaly@gmail.com | (650) 586-9720 | linkedin.com/in/douglasdaly | github.com/dougdaly

SUMMARY

Principal AI Engineer and Technical Leader with 20+ years of experience building systems that help organizations understand what is happening, why it is happening, and what to do next. Expertise spans operational intelligence, decision-support systems, attribution analytics, observability, graph analytics, and AI-enabled business systems.

Designed and delivered solutions ranging from a patented radar resource-management system and enterprise-scale operational monitoring platforms to merchant intelligence systems, reliability analytics, graph-based dependency modeling, GenAI assistants, and AI-enabled decision support solutions. Combines deep expertise in machine learning, cloud platforms, data engineering, and modern LLM architectures with a proven ability to translate complex business problems into practical, measurable outcomes.

CORE TECHNOLOGIES

AI & Machine Learning: LLMs | RAG | Agentic AI | LangGraph | CrewAI | AutoGen | MCP | Prompt Engineering | PEFT/LoRA | Fine-Tuning | Diffusion Models | CLIP | Scikit-Learn | XGBoost

Cloud & Data Platforms: AWS | GCP | Vertex AI | BigQuery | SageMaker | Bedrock | S3 | Neptune

Programming & Analytics: Python | SQL | Tableau | Pandas | NumPy | Jupyter | Elastic Stack | Splunk | Kafka

Data & Systems Engineering: Graph Analytics | Observability | Operational Intelligence | Entity Resolution | Data Modeling | Forecasting | Attribution Analytics

PROFESSIONAL EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

- {'label': 'Consolidated Edison (via ABC Consulting)', 'type': 'bullet', 'content': ['Led analytical readiness assessment for Oracle/EBS and Maximo data migrated to BigQuery, establishing the foundation for cross-domain analytics across construction operations and financial systems.', 'Developed automated profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation frameworks across 100+ source tables and millions of operational and financial records.', 'Developed a phased analytical maturity roadmap adopted as the foundation for anomaly detection, construction analytics, and AI-enabled decision support initiatives.', 'Built a governed analytics assistant using Gemini, BigQuery, and tool-calling architectures that applied analyst rules and operational controls while maintaining predictable and auditable execution.']}
- {'label': 'NBCUniversal (via Apex Systems)', 'type': 'bullet', 'content': ['Built AWS-based AI and graph analytics pipelines using SageMaker, Terraform, S3, and containerized workloads.', 'Developed graph-ready service dependency datasets from API telemetry and operational metrics to support digital twin initiatives.', 'Evaluated graph analytics and graph machine learning approaches using Neptune, GraphStorm, Bedrock, and LangGraph for dependency analysis and operational intelligence.', 'Authored architecture and analytical readiness assessments identifying data and operational gaps affecting long-term digital twin objectives.']}

AI Instructor | SimpliLearn | October 2025 - Present

- Delivered professional AI training across multi-month cohorts for software engineers, data scientists, and technical professionals through university- and enterprise-sponsored continuing education programs.
- Taught transformer architectures, LLMs, RAG, agentic AI, multimodal AI, MCP, diffusion models, and enterprise AI governance.
- Developed hands-on demonstrations, Jupyter notebooks, and end-to-end applications covering modern AI workflows, evaluation techniques, and deployment considerations.
- Mentored students on AI architecture, governance, implementation tradeoffs, and enterprise adoption strategies.

AI Instructor | Interview Kickstart | March 2025 - Present

- Delivered AI and machine learning instruction for software engineering professionals pursuing advanced technical roles.
- Taught practical applications of LLMs, RAG, agentic AI, and modern AI development workflows.
- Mentored students on AI architecture, implementation patterns, and technology selection tradeoffs.

Data Scientist | Intuit (via The Cydio Group) | Apr 2025 - Jul 2025

- Delivered analytics supporting business and operational decision-making for QuickBooks Capital.
- Investigated attribution anomalies across customer acquisition funnels, marketing campaigns, and conversion pathways to improve measurement accuracy and identify emerging business trends.
- Consolidated fragmented reporting into unified Tableau dashboards and automated executive reporting workflows using Python and SQL, reducing report preparation from days to hours.
- Partnered with finance, marketing, and product stakeholders to translate attribution and performance data into actionable business recommendations.

Data Scientist | Meta | October 2021 - January 2025

- Defined reliability KPIs and measurement frameworks for Meta FinTech, establishing leading indicators used to guide reliability planning, prioritization, and performance management.
- Developed forecasting and performance-management frameworks across business pillars and sub-pillars, translating engineering initiatives into projected reliability outcomes used for planning, goal setting, and leadership decision-making.
- Built attribution frameworks linking incidents, alerts, defects, and operational metrics to engineering organizations and service ownership structures, enabling more accurate accountability and performance measurement.
- Designed and developed the ReMon framework, enabling engineering teams to build comprehensive monitoring and alerting systems without requiring expertise in complex internal tooling; adopted across Meta FinTech, contributing to the automatic detection of more than 15 SEVs in 2024.

- Delivered impact sizing, ROI, and risk exposure analyses supporting investment and prioritization decisions across platform reliability, privacy compliance, authentication, testing infrastructure, and data platform initiatives.

Data Scientist | Samsung Electronics (via Harvey Nash) | Dec 2020 - Oct 2021

- Developed and deployed gradient-boosting propensity models used to target digital marketing campaigns, increasing campaign conversions by 146% and campaign-attributed revenue by 229%.
- Built customer sentiment analysis pipelines using NLTK and survey data to identify product and customer experience insights.
- Automated audience segmentation and campaign targeting workflows, improving efficiency and scalability across digital marketing initiatives.
- Improved marketing attribution and performance measurement capabilities through predictive modeling and analytics-driven decision support.

Head of Data Science | Facteus | Oct 2017 - Nov 2020

- Led data science initiatives on a 30B+ transaction dataset, creating analytical products and market-intelligence capabilities through large-scale feature engineering, merchant intelligence, and consumer-spending analytics.
- Developed merchant, brand, industry, and storefront resolution methodologies that transformed cryptic payment-card transactions into business-identifiable entities, enabling accurate market intelligence across more than 1,700 businesses with greater than 99% transaction coverage.
- Evaluated prospective data partnerships by assessing signal quality, coverage, and business value, supporting strategic investment and acquisition decisions.
- Produced transaction-based industry analyses during the COVID-19 pandemic, identifying shifts in consumer behavior, regional economic impacts, and e-commerce adoption trends before traditional reporting sources became available.
- Created data-driven reports and visualizations used by hedge funds, partners, and internal stakeholders to understand emerging market trends and business performance.

Expert SEO Solutions Architect | Nike (via Intersoft Inc.) | Nov 2016 - Sep 2017

- Designed a scalable SEO and information-architecture framework for Nike's product discovery experience, balancing customer navigation paths with search-engine indexation requirements across a rapidly expanding product catalog.
- Developed canonical URL, redirect, and XML sitemap strategies that improved search visibility while preventing duplicate content, redirect conflicts, and circular routing issues.
- Built Node.js proof-of-concept solutions demonstrating the feasibility and effectiveness of proposed SEO approaches, supporting adoption by business and engineering teams.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Developed predictive capacity-management models for customer-facing banking platforms, forecasting peak demand scenarios using customer-demand, temporal, and business-process drivers to support infrastructure planning and risk management.
- Identified critical gaps in enterprise monitoring operations after discovering that major incidents were frequently detected by customer call channels rather than operational monitoring systems.
- Conceived, designed, and led adoption of RITHM (Real-Time IT Health Monitoring), an operational intelligence platform that aggregated monitoring signals by business capability and system ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship for enterprise monitoring modernization, expanding initial Splunk-based prototypes into a scalable Kafka and Elastic Stack solution aligned with Capital One's open-source strategy.
- Partnered with CTO-level leadership to define enterprise monitoring strategy, operational health metrics, and reliability-improvement initiatives.
- Awarded the Capital One Business Innovation Award for contributions to operational intelligence and enterprise monitoring transformation; presented work at internal and external industry conferences.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering organizations of 40-50 engineers specializing in modeling, simulation, analysis, and advanced aerospace systems, providing technical leadership, career development, and proposal support for programs valued up to \$50M.
- Turned around an over-budget classified fixed-price technology program at approximately one-third completion by establishing execution plans, aligning engineering teams, and proactively managing technical and program risks; delivered the program under budget and nearly doubled the original profit target on a historically challenging delivery effort.
- Served as Principal Investigator for advanced signal-generation and radar resource-management algorithms, leading research, simulation, flight testing, and operational deployment across multiple tactical and surveillance aircraft programs.
- Developed a patented fuzzy-logic resource-management system (U.S. Patent, 2014) resulting from radar scheduling innovations deployed on major Department of Defense aircraft platforms.

SELECTED PROJECTS

GraphRAG Engine & LoRA Coverage Assistant | GitHub | 2025

- Built a GraphRAG system for multi-hop benefits reasoning with fully traceable retrieval paths across entities, clauses, and supporting evidence.
- Fine-tuned a LoRA adapter on a 7B open-source LLM to enforce a strict JSON schema (decision/reason/disclaimer) for reliable downstream automation.

Agent-to-Agent Protocol MVPs | GitHub | 2026

- Agent Olympics: built a small evaluation runner that benchmarks a generalist agent vs discovered specialists across three task categories with explicit scoring criteria and summary reporting.
- AI PEMDAS: built an instrumented coordination workflow where an evaluator routes sub-steps to ADD vs MULT agents and logs calls for end-to-end traceability (FastAPI JSON-RPC).

Agentic Multimodal Orchestrator | GitHub | 2025

- Orchestrated structured retrieval, layout planning, and diffusion prompting to generate data-driven posters/maps with repeatable, template-based workflows.

- Implemented template constraints and parameterized prompts to ensure consistency across runs.

EDUCATION

MBA | (UCLA Anderson School of Management): Top Honors | GPA: 3.9

M.S., Applied Statistics | (Rochester Institute of Technology): GPA: 3.9

M.S., Electrical Engineering | (University of Washington): GPA: 3.7

B.S., Physics (Minor: Mathematics) | (Humboldt State University): GPA: 3.4

PATENTS & RECOGNITION

- Lead Inventor, U.S. Patent #8635622: Method and System for Resource Management Using Fuzzy Logic Timeline Filling (2014)
- Capital One Business Technology Innovation Award (2014)
- Conference Chair & Speaker - Customer Analytics Innovation Summit (2016)

CERTIFICATIONS

- Google Associate Data Practitioner
- Google Ads Search Certification (Expired)
- Google Analytics GA4 Certification (Expired)